

Master's Thesis Progress Documentation

Phase 6: Interactive Visual Analytics Dashboard in Power BI

1. Thesis Title

Design and Evaluation of an Interactive Visual Analytics System for Explainable Artificial Intelligence

2. Purpose of This Document

This document summarizes the work completed during Phase 6 of the master's thesis. The main purpose of this phase was to transform the machine-learning predictions and SHAP explanations produced in the previous phases into a user-facing interactive visual analytics system in Microsoft Power BI.

The dashboard was designed for non-technical users who need to identify customers at risk of churn, understand the strength of the prediction, investigate the model's global behaviour, inspect customer-specific explanations, compare customer segments, and review model performance.

Power BI was used only for visualization, interaction, filtering, and analytical presentation. Model training, probability generation, threshold selection, and SHAP calculation remained in Python. This separation ensured that the dashboard displayed validated analytical outputs without recalculating the machine-learning or explainability results inside Power BI.

3. Current Stage of the Thesis

The thesis has now completed the complete technical implementation pipeline:

- Research foundation and planning
- Dataset selection and exploratory data analysis
- Data cleaning and preprocessing
- Machine-learning model development
- Explainable Artificial Intelligence analysis using SHAP
- Interactive visual analytics dashboard development in Power BI

The completed technical pipeline is:

Customer Data -> Preprocessing -> Gradient Boosting Model -> Predictions and Probabilities -> SHAP Explanations -> Power BI Dashboard

4. Main Objective of Phase 6

The objective of Phase 6 was to create an interactive dashboard that supports prediction interpretation and business-oriented decision making. The system was required to present both predictive results and explainability information in a consistent interface.

The dashboard was designed to answer the following practical questions:

- Which customers have the highest predicted churn risk?
- How high is the churn probability for a selected customer?
- Was the customer predicted to churn or remain?
- Which features increased or reduced the customer's predicted churn probability?

- Which original customer features are most influential across the complete test dataset?
- How do customer segments differ in churn probability and churn rate?
- How did the selected model and threshold perform?
- What was gained by lowering the classification threshold from 0.50 to 0.27?

5. Power BI Environment and Input Data

The dashboard was developed in Microsoft Power BI Desktop in a Windows environment. The final report contains six connected report pages with a common header, consistent page navigation, reusable visual formatting, interactive slicers, dynamic measures, and customer-level filtering.

A dedicated Power BI import package was prepared from the Phase 4 and Phase 5 outputs. The package contained the following eight CSV files:

Input file	Purpose in Power BI
customer_prediction_explanations.csv	Main customer-level table containing original attributes, actual churn, predicted churn, churn probability, threshold, SHAP summary fields, and customer identifiers.
shap_values_grouped_long.csv	Long-format grouped SHAP table containing one row per customer and original feature.
customer_top_factors_combined.csv	Top positive and negative customer-level factors with rank, value, direction, and SHAP contribution.
global_shap_importance_grouped.csv	Global importance and average signed contribution for the 19 grouped original features.
final_test_metrics.csv	Final evaluation metrics for the selected model and threshold comparison.
confusion_matrix_results.csv	True-positive, false-positive, false-negative, and true-negative results.
model_comparison_results.csv	Cross-validation results for the evaluated machine-learning models.
threshold_analysis.csv	Performance values across threshold configurations, including the default and selected thresholds.

6. Power BI Data Model

The customer-level table was used as the central table because it contains one unique row for each of the 1,407 test customers. The customer identifier was preserved as text and used as the relationship key.

The main relationships were created as follows:

- Customers[customerID] 1 -> many Customer SHAP[customerID]
- Customers[customerID] 1 -> many Customer Top Factors[customerID]

The global SHAP, final-metric, confusion-matrix, threshold-analysis, and model-comparison tables were kept disconnected where appropriate because they represent model-wide results rather than customer-level records. Filtering was normally propagated in one direction from the Customers table to the customer-level explanation tables.

The data model was validated to prevent many-to-many relationships and duplicate-customer totals. Customer counts and churn counts were always calculated from the one-row-per-customer table rather than from long-format SHAP tables.

7. Measures and Supporting Structures

A dedicated Dashboard Measures table was created to store the reusable DAX calculations. The measures were organized around customer counts, risk analysis, customer selection, model performance, and threshold comparison.

The main measures and calculated structures included:

- Total Customers, Actual Churn Customers, Predicted Churn Customers, and Average Churn Probability
- High-Risk Customers, Medium-Risk Customers, and Low-Risk Customers
- Selected Threshold, Recall, F1-score, ROC-AUC, Precision, Accuracy, PR-AUC, and Brier Score
- Selected Customer ID, Selected Churn Probability, Selected Predicted Churn, Selected Actual Churn, Selected Customer Base Value, and Risk Margin versus Threshold
- Dynamic selected-customer explanation text
- Highest-risk segment measures for Contract, Internet Service, and Payment Method
- Segment comparison measures and disconnected axis tables
- Default and selected false-negative measures and a waterfall calculation
- Probability bands, risk categories, customer profile field tables, and metric-axis helper tables

The selected classification threshold remained 0.27. It was used to convert churn probability into the final churn or non-churn decision. It did not alter the SHAP contributions, which remained on the probability scale.

8. Dashboard Page Development

The final report contains six pages. A common page-navigation bar was placed at the top of every page so that users can move directly between the dashboard sections. The page tabs also remain available at the bottom of the Power BI report.

8.1 Executive Overview

- Provides the main model and customer-risk summary for the complete test dataset.
- Displays 1,407 customers, 374 actual churners, 609 predicted churners, a 27.2% average churn probability, a 0.27 threshold, 80.75% recall, 0.8383 ROC-AUC, and a 0.6144 F1-score.
- Includes actual and predicted churn distributions, probability bands, actual-versus-predicted counts, a confusion matrix, and key interpretation statements.

8.2 Customer Risk Analysis

- Supports interactive identification and prioritization of high-risk customers.
- Provides slicers for contract, internet service, payment method, predicted churn, actual churn, and risk category.
- Includes risk-category cards, the top high-risk customer ranking, a risk-distribution chart, a tenure-versus-probability scatter plot, a detailed customer-risk table, and analyst guidance.

8.3 Global Model Explanation

- Explains model behaviour across all 1,407 test customers using grouped original features.
- Shows global mean absolute SHAP importance, average signed contribution, and a ranked Top 10 feature table.
- Includes relationship views for Tenure and Monthly Charges and an interpretation panel that explicitly avoids causal claims.

8.4 Individual Customer Explanation

- Provides a dynamic local explanation for one selected customer.
- Displays the selected customer profile, churn probability, predicted and actual churn labels, threshold, SHAP base value, and risk margin.
- Includes a probability gauge, factors increasing risk, factors reducing risk, and a dynamic natural-language explanation that updates with the customer selector.

8.5 Customer Segment Comparison

- Supports comparison across contract, internet service, payment method, senior-citizen status, and paperless billing.
- Shows highest-risk segment KPIs, four segment comparison charts, a segment summary matrix, and an overall interpretation panel.
- Enables users to study segment-level differences through interactive dropdown slicers.

8.6 Model Performance Summary

- Summarizes the selected model and explains the threshold decision.
- Displays Accuracy 73.06%, Precision 49.59%, Recall 80.75%, F1-score 0.6144, ROC-AUC 0.8383, PR-AUC 0.6520, Brier Score 0.1386, and threshold 0.27.
- Includes a ribbon chart for model ranking across metrics, a default-versus-selected threshold comparison, a false-negative reduction waterfall, and a threshold-rationale panel.

9. Interactivity and User-Facing Functionality

The interactive system was designed to support practical evaluation tasks. The implemented interactions include:

- Page-to-page navigation using a consistent navigation bar
- Dropdown slicers for customer and segment filtering
- A single-customer selector for local explanation analysis
- Dynamic KPI cards, profile values, SHAP charts, gauge values, and explanatory text
- Cross-filtering between slicers and customer-risk visuals
- Interactive customer tables for reviewing high-risk cases
- Segment-level comparisons for contract, service, payment, demographic, and billing groups
- Visual comparison of default and selected classification thresholds

The interactivity allows a participant to move from a general model overview to a specific customer explanation and then compare the result with segment-level and model-level information.

10. Dashboard Design and Interpretation Rules

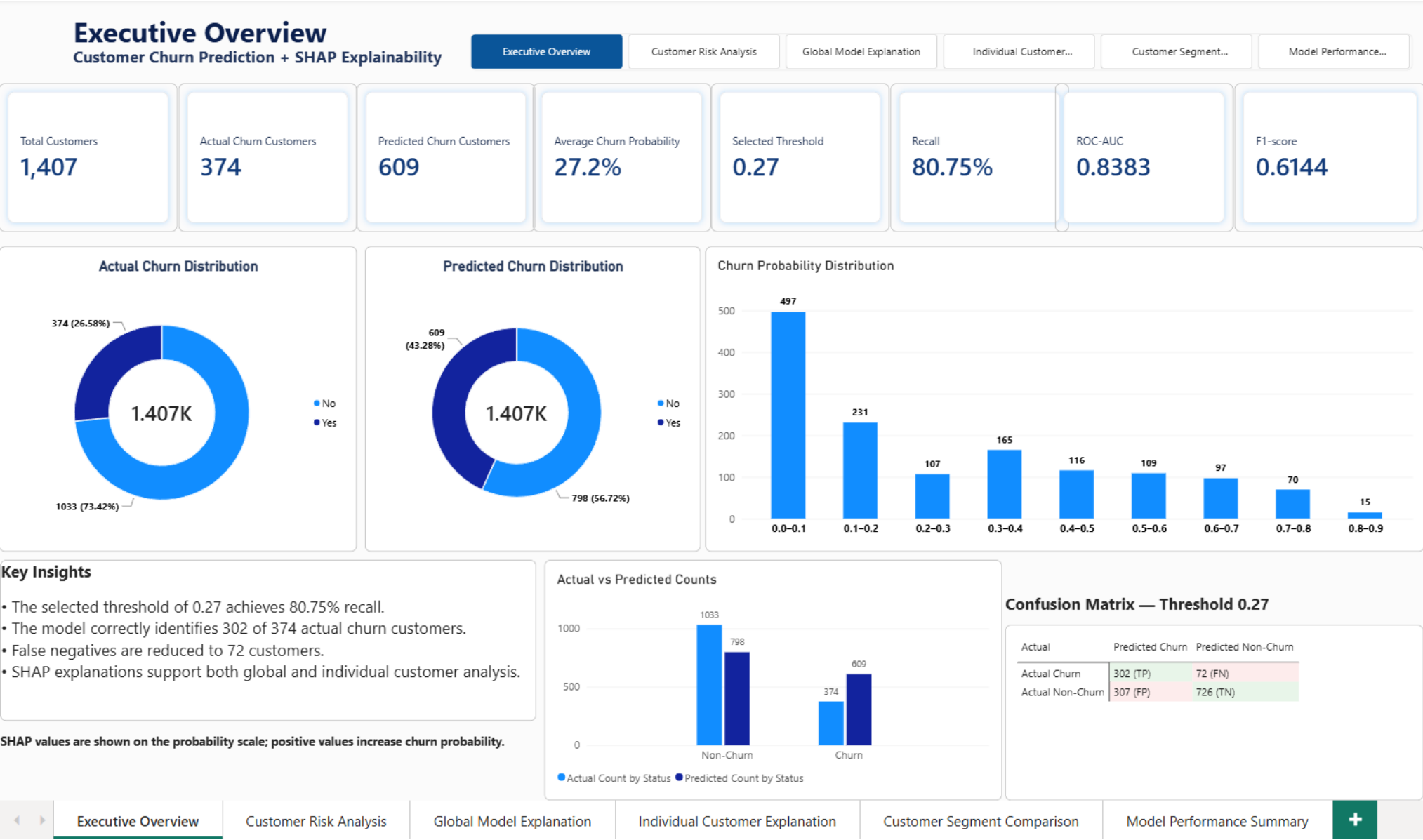
The report was designed with a consistent white background, dark-blue headings, blue and orange analytical accents, rounded borders, equal card dimensions, and aligned visual containers. The same navigation order and page titles are used throughout the report.

The following interpretation rules were applied:

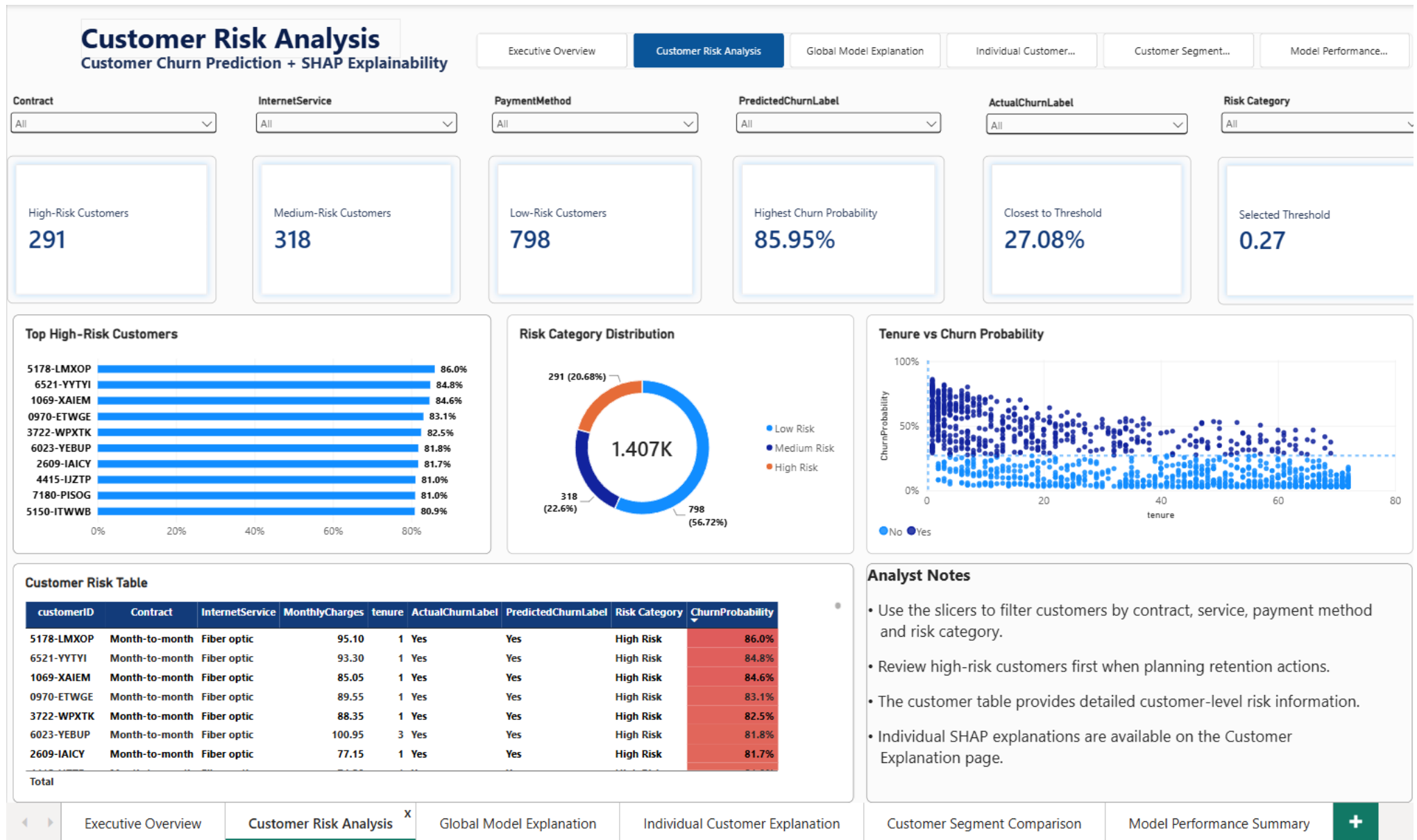
- Original feature names are shown instead of technical one-hot encoded feature names.
- Original customer values are shown instead of standardized model inputs.
- Actual churn and predicted churn are displayed as separate fields.
- Positive SHAP values are described as increasing the predicted churn probability.
- Negative SHAP values are described as reducing the predicted churn probability.

- SHAP values are described as model explanations and associations, not proof of causation.
- The threshold is presented as a decision rule applied after probability generation.
- Counts are calculated from the customer-level table to prevent duplicate totals.

8.1 Executive Overview



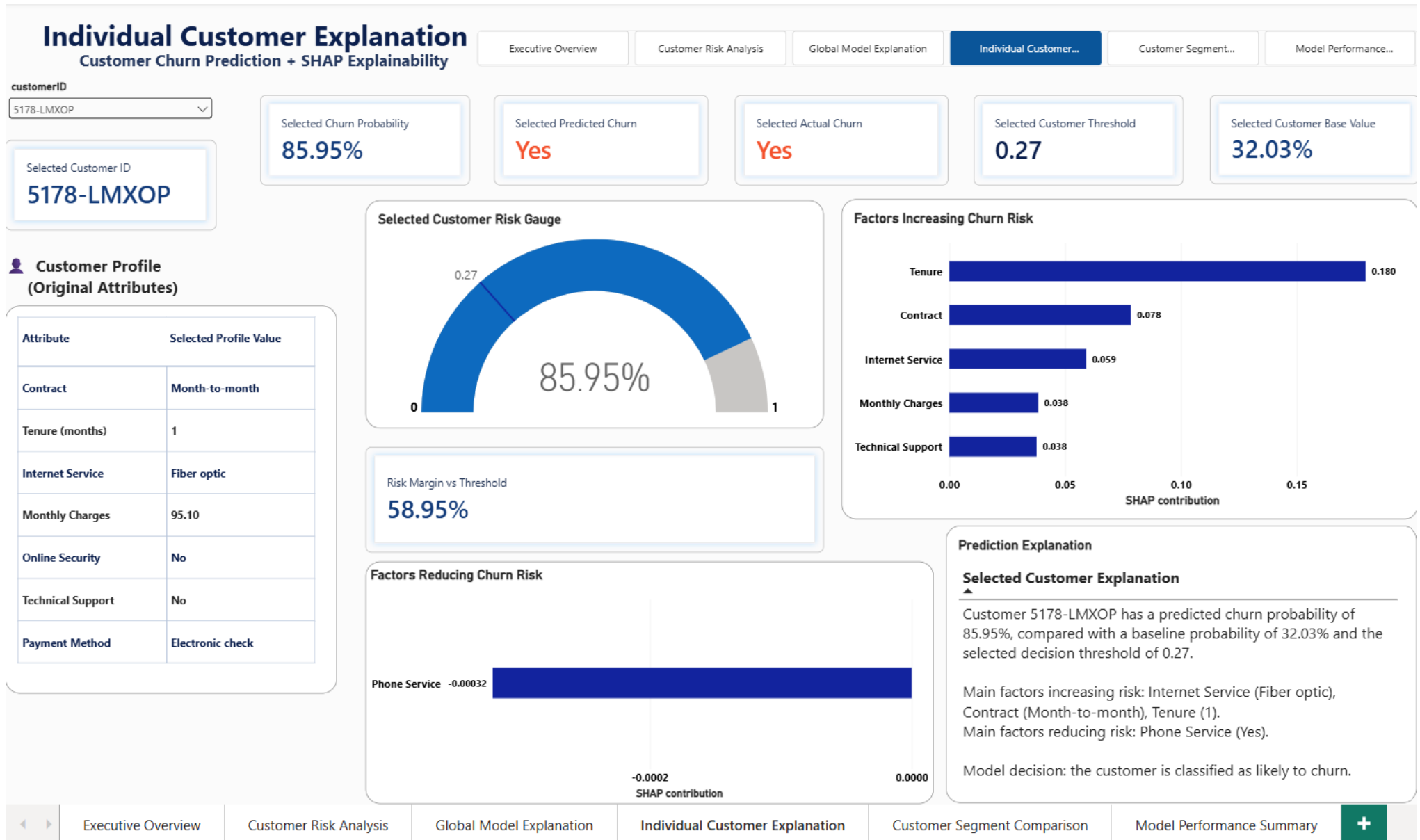
8.2 Customer Risk Analysis



8.3 Global Model Explanation



8.4 Individual Customer Explanation



Customer Profile (Original Attributes)

Attribute	Selected Profile Value
Contract	Month-to-month
Tenure (months)	1
Internet Service	Fiber optic
Monthly Charges	95.10
Online Security	No
Technical Support	No
Payment Method	Electronic check

Selected Customer Risk Gauge

Risk Margin vs Threshold

58.95%

Factors Increasing Churn Risk

Factor	SHAP contribution
Tenure	0.180
Contract	0.078
Internet Service	0.059
Monthly Charges	0.038
Technical Support	0.038

Factors Reducing Churn Risk

Factor	SHAP contribution
Phone Service	-0.00032
	-0.0002
	0.0000

Prediction Explanation

Selected Customer Explanation

Customer 5178-LMXOP has a predicted churn probability of 85.95%, compared with a baseline probability of 32.03% and the selected decision threshold of 0.27.

Main factors increasing risk: Internet Service (Fiber optic), Contract (Month-to-month), Tenure (1).

Main factors reducing risk: Phone Service (Yes).

Model decision: the customer is classified as likely to churn.

Executive Overview

Customer Risk Analysis

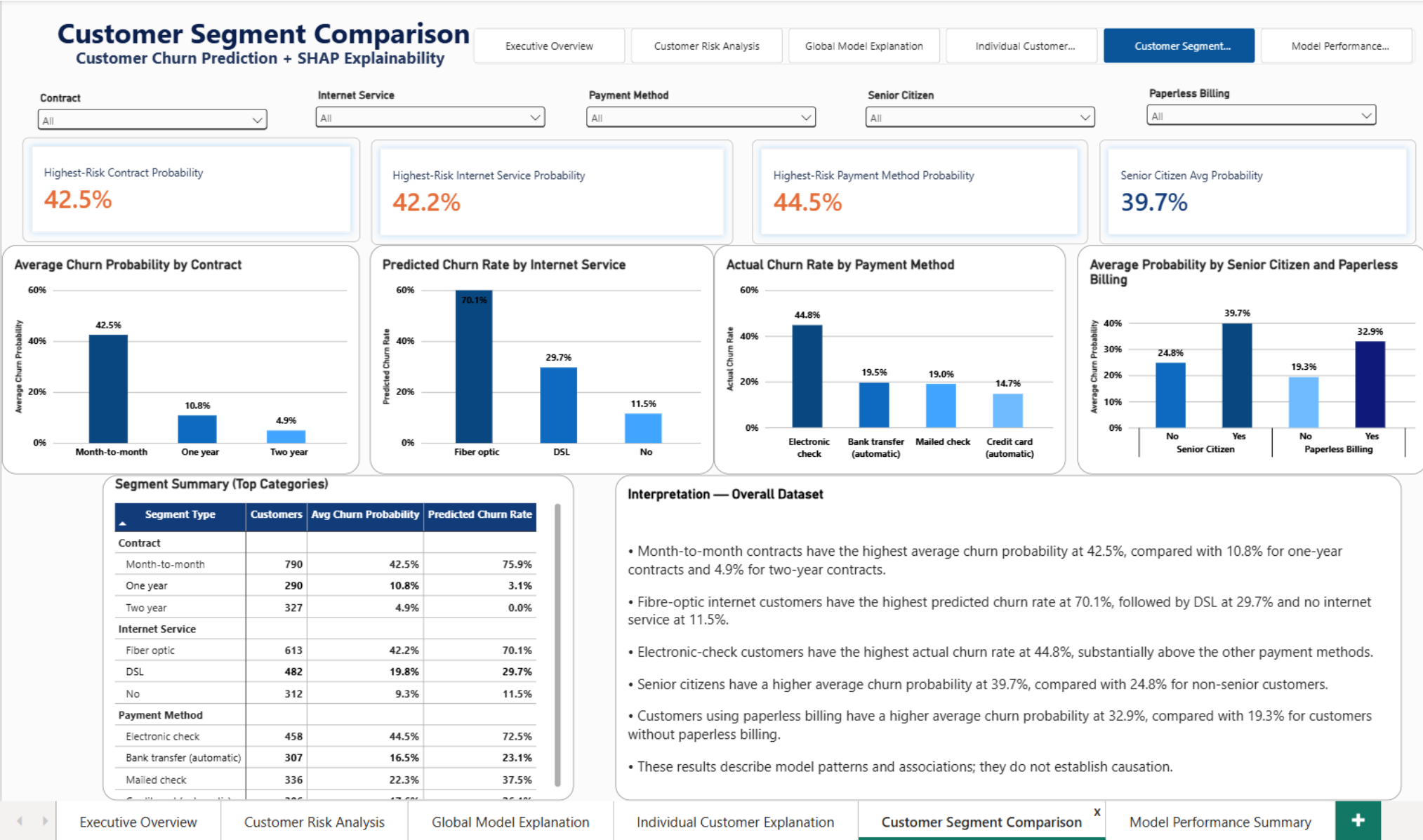
Global Model Explanation

Individual Customer Explanation

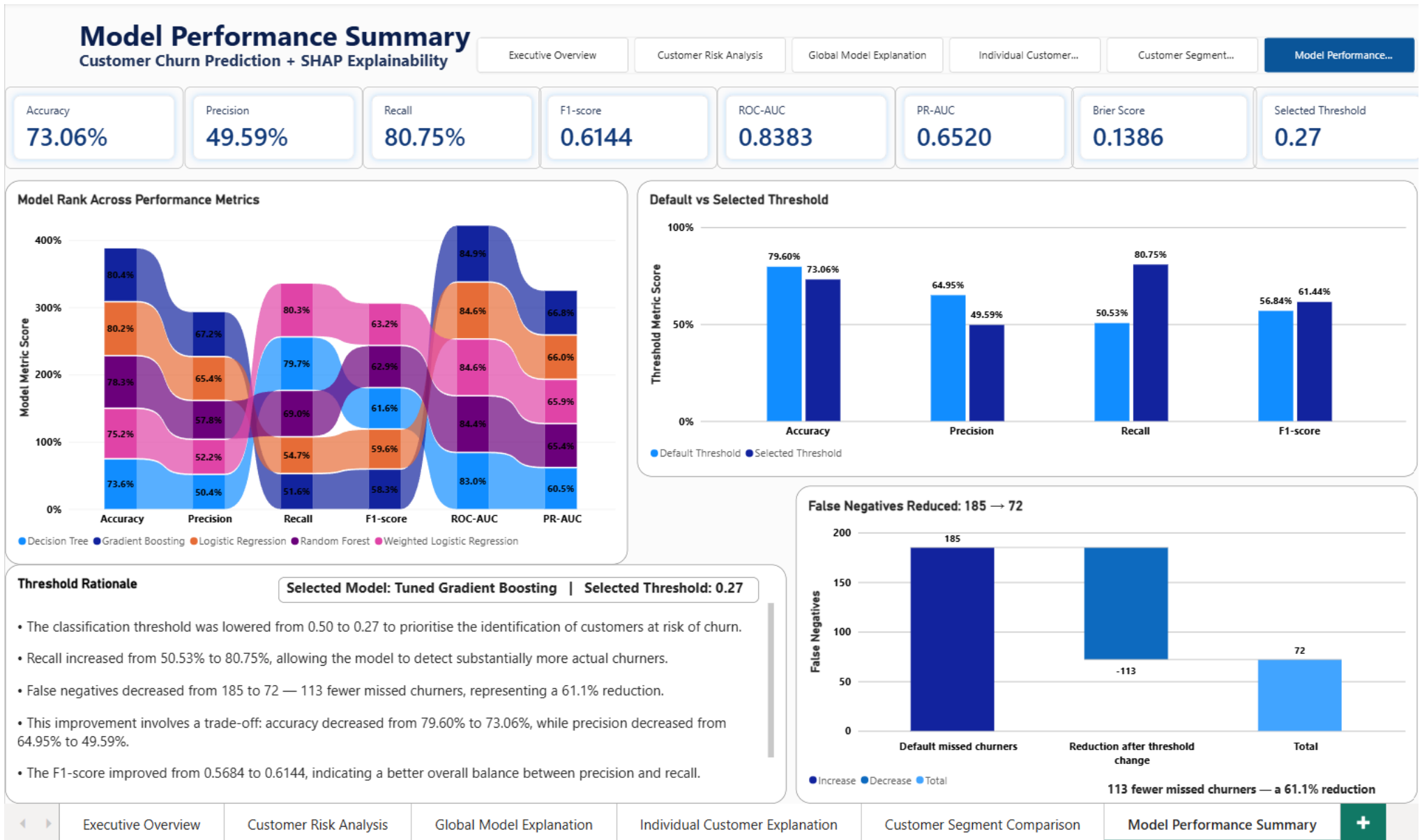
Customer Segment Comparison

Model Performance Summary

8.5 Customer Segment Comparison



8.6 Model Performance Summary



11. Validation and Quality Checks

The dashboard was reviewed page by page during development. Screenshots were checked after every major construction step, and errors were corrected before continuing to the next section.

The final validation included the following checks:

- The customer table contains 1,407 unique test customers.
- The customer slicer correctly updates the selected-customer cards, profile, gauge, SHAP charts, and dynamic explanation.
- The risk-category counts sum to 1,407 customers: 291 high-risk, 318 medium-risk, and 798 low-risk customers.
- The dashboard displays 374 actual churn customers and 609 predicted churn customers.
- The selected threshold of 0.27 is displayed consistently across the relevant pages.
- The selected customer 5178-LMXOP displays an 85.95% churn probability and the expected positive and negative SHAP factors.
- Global SHAP views show the 19 grouped original features and the validated importance ranking.
- Segment charts and matrices respond to the relevant slicers without duplicate customer totals.
- Model-performance cards match the validated Phase 4 results.
- The threshold comparison correctly shows recall increasing from 50.53% to 80.75%.
- False negatives decrease from 185 to 72, a reduction of 113 customers or approximately 61.1%.
- All six pages contain working navigation and consistent visual formatting.

12. Main Analytical Results Presented in the Dashboard

Result	Value
Selected model	Tuned Gradient Boosting Classifier
Test customers	1,407
Actual churn customers	374
Predicted churn customers	609
Selected threshold	0.27
Accuracy	73.06%
Precision	49.59%
Recall	80.75%
F1-score	0.6144
ROC-AUC	0.8383
PR-AUC	0.6520
Brier Score	0.1386
True positives	302
False negatives	72
False positives	307
True negatives	726
SHAP output scale	Probability
SHAP base value	Approximately 0.3203
Grouped original features	19

13. Phase 6 Deliverables

The following deliverables were produced or prepared during Phase 6:

- Interactive six-page Power BI dashboard (.pbix)
- Phase_6_PowerBI_Import.zip containing the selected Power BI-ready CSV files

- Six final dashboard screenshots for documentation and static reporting
- Phase 6 progress documentation PDF
- Updated GitHub repository structure and README information

Recommended repository placement:

- documentation/Phase_6_Interactive_Visual_Analytics_Dashboard.pdf
- dashboards/Phase_6_Interactive_XAI_Dashboard.pbix
- outputs/Phase_6_PowerBI_Import.zip
- screenshots/Phase_6_Dashboard_Pages/

14. Completion Status and Next Steps

Phase 6 is complete. The complete machine-learning, explainability, and interactive visualization system has now been implemented.

The remaining thesis work will focus on:

1. Creating a static explanation baseline containing comparable information without interactive functionality.
2. Designing the user-evaluation protocol and participant tasks.
3. Comparing the static baseline with the interactive Power BI dashboard.
4. Measuring interpretation accuracy, completion time, confidence, trust, and decision quality.
5. Analysing the evaluation results.
6. Writing and finalizing the master's thesis.

15. Conclusion

Phase 6 successfully transformed the prediction and SHAP outputs into an interactive visual analytics system. The final dashboard combines customer-level risk assessment, global model interpretation, individual explanation, segment comparison, and model-performance analysis within one consistent Power BI report.

The dashboard enables users to move from an overall view of churn risk to detailed customer and feature-level explanations. It also makes the selected threshold and its business trade-offs visible. This completes the technical implementation of the proposed system and provides the interactive interface required for the planned comparison with a static explanation baseline.